

Distributed robust consensus of linear multi-agent systems with switching topologies

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Abstract: This paper studies the consensus problem of multi-agent systems consisting of general linear node dynamics with external disturbances in transmission channels under fixed and switching communication topologies. A distributed observer-type robust consensus protocol based on relative output measurements is proposed. A model transformation approach is introduced to address the robust consensus of multi-agent systems. Some conditions for robust consensus are given in terms of linear matrix inequalities for fixed and jointly connected communication topologies. A multi-step robust consensus protocol design procedure is further presented. Finally, the effectiveness of the theoretical results is demonstrated through numerical simulations.

1 Introduction

Recently, distributed coordinated control of multi-agent systems has attracted a great deal of attention from various scientific communities because of its broad applications in such areas as satellite formation flying, cooperative unmanned air vehicles, scheduling of automated highway systems, air traffic control and many other industries. Multi-agent systems are defined as agents forming a network in which information is exchanged among these agents through a communication network. A critical problem for multi-agent systems is to develop appropriate distributed control policies for each agent, using local information from its neighbours, such that the networked systems can achieve an agreement on certain quantities of interest. This problem is usually called the consensus problem.

The consensus problem [1–13] has been extensively studied by numerous researchers from various perspectives. In [14], a notion of an H_∞ consensus performance region was proposed to address robust consensus. In [15], consensus based on an observer for both continuous and discrete time linear multi-agent systems was addressed. In [16], a simple model for phase transition of a group of self-driven particles was proposed. In [17], a theoretical explanation for the behaviour observed in a simple model was provided by using the graph theory. In [18], a general framework of the consensus problem for networks of dynamic agents with fixed or switching topologies and communication time delays was established. In [19], the consensus under dynamical interaction topologies was studied. In [20], some necessary and sufficient conditions for second-order consensus were given. In [21], tracking control for multi-agent consensus with an active leader was studied and a local controller together with a neighbour-based state-estimation rule was designed. In [22], the consensus problem for directed networks of agents with external disturbances and model uncertainties on fixed and switching topologies was investigated.

Much attention has been paid to consensus of first-order or second-order integrator dynamics. However, networked systems with general linear systems are more interesting and include the integrator dynamics of any order as a special case. In many applications, full state information is not always available for controller design, so output feedback design is required. In practical networked multi-agent systems, various external disturbances are unavoidable in information acquisition and transmission that may cause the networked systems to diverge. Hence, the external disturbances should be taken into consideration when designing a consensus protocol.

This paper is concerned with the robust consensus problem of multi-agent systems with external disturbances in transmission

channels under fixed and switching communication topologies, where each agent has general linear dynamics. A distributed observer-type robust consensus protocol based on relative output measurement is proposed. A model transformation approach based on the properties of the reduced Laplacian matrix is introduced to address the robust consensus of multi-agent systems. Some conditions for robust consensus in terms of linear matrix inequalities and the appropriate solutions are given for both fixed and switching communication topologies. A multi-step robust consensus protocol design procedure is further presented. The design technique is mainly based on algebraic graph theory [23], matrix inequality [24] and linear control theory.

The rest of this paper is organised as follows. Section 2 introduces some basic notations and reviews some useful results of algebraic graph theory. Section 3 presents a distributed observer-type robust consensus protocol for multi-agent systems. Numerical examples are simulated to verify the theoretical analysis in Section 4. Section 5 concludes this paper.

2 Preliminaries

In this section, some basic notations and useful results about algebraic graph theory that are related to our later analysis are briefly summarised.

Let $\mathbf{g} = (V, \varepsilon, A)$ be a weighted directed graph of order N , with a node set $V = [v_1, v_2, \dots, v_N]$, a directed edge set $\varepsilon \subseteq v \times v$ and a weighted adjacency matrix $A = [a_{ij}] \in R^{N \times N}$. An edge (v_i, v_j) in a weighted directed graph denotes that agent j can obtain information from agent i , but not necessarily vice versa. The weighted adjacency matrix A of a weighted directed graph is defined such that $a_{ii} = 0$, a_{ij} is a positive weight if $(v_j, v_i) \in \varepsilon$ and $a_{ij} = 0$ if $(v_j, v_i) \notin \varepsilon$. In this paper, only positively weighted directed graphs are considered, that is, $a_{ij} > 0$ if and only if there is a directed edge (v_j, v_i) in graph $\mathbf{g}$.

A directed path is a sequence of edges in a directed graph of the form $(v_{i_1}, v_{i_2}), (v_{i_2}, v_{i_3}), \dots$, where $v_{i_j} \in V$. A directed graph has a directed spanning tree if there exists at least one node having a directed path to all of the other nodes. A root r is a node having the property that for each node v different from r , there is a directed path from r to v .

Let the Laplacian matrix $L = [l_{ij}] \in R^{N \times N}$ associated with the adjacency matrix A be defined as

$$l_{ij} = \begin{cases} l_{ij} = -a_{ij}, & i \neq j \\ \sum_{j=1}^N a_{ij}, & i = j \end{cases} \quad (1)$$

The following lemma [19] shows some basic properties of the Laplacian matrix $\mathbf{L}$.

Lemma 1: Let $\mathbf{L} \in \mathbb{R}^{N \times N}$ be the Laplacian matrix of a directed graph $\mathbf{g}$ and $\mathbf{1} = [1, 1, \dots, 1]^T \in \mathbb{R}^{N \times 1}$, then

1. $\mathbf{L}$ has at least one zero eigenvalue, and $\mathbf{1}$ is the associated eigenvector, that is $\mathbf{L}\mathbf{1} = \mathbf{0}$.

2. If $\mathbf{g}$ has a spanning tree, then zero is a simple eigenvalue of $\mathbf{L}$, and all the other $N - 1$ eigenvalues have positive real parts.

Introduce the following matrix

$$\tilde{\mathbf{L}} = \begin{bmatrix} l_{22} - l_{12} & \cdots & l_{2n} - l_{1n} \\ \vdots & \ddots & \vdots \\ l_{n2} - l_{12} & \cdots & l_{nn} - l_{1n} \end{bmatrix} \quad (2)$$

where l_{ij} is the entry of the Laplacian matrix $\mathbf{L}$. $\tilde{\mathbf{L}}$ is referred to as the reduced Laplacian matrix.

Lemma 2: The eigenvalues of the reduced Laplacian matrix $\tilde{\mathbf{L}}$ consist of the rest of the eigenvalues of Laplacian matrix $\mathbf{L}$, except a zero eigenvalue. If $\mathbf{g}$ has a spanning tree, all eigenvalues of $\tilde{\mathbf{L}}$ have positive real parts.

Proof: There exists a similarity transformation between the Laplacian matrix $\mathbf{L}$ and Δ , that is, $\Delta = \mathbf{U}^{-1}\mathbf{L}\mathbf{U}$, where

$$\Delta = \begin{bmatrix} 0 & \beta \\ 0 & \tilde{\mathbf{L}} \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} 1 & 0 \\ \mathbf{1}_{n-1} & \mathbf{I}_{n-1} \end{bmatrix}$$

According to Lemma 1, Lemma 2 can be obtained easily.

To describe the variable communication topology, a switching signal $\sigma(t): [0, +\infty) \rightarrow \{1, \dots, n\}$ is introduced whose value at time t is the index of the graph at time t . Consider an infinite sequence of continuous, bounded, non-overlapping time intervals $[t_k, t_{k+1})$, $k = 0, 1, 2, \dots$ with $t_0 = 0$ and $t_{k+1} - t_k \leq T$ for some constant $T > 0$. In each interval $[t_k, t_{k+1})$, there is a sequence of subintervals

$$[t_k^0, t_k^1), \dots, [t_k^{j-1}, t_k^j), \dots, [t_k^{m_k-1}, t_k^{m_k})$$

with t_k^0, t_k and $t_k^{m_k} = t_{k+1}$ for some integer m_k . The communication topology switches at times t_k^{j-1} that satisfy $t_k^j - t_k^{j-1} \geq T_0$ and $1 \leq j \leq m_k$. The communication topology does not change during each subinterval $[t_k^{j-1}, t_k^j)$.

For a symmetric matrix $\mathbf{P}$, by $\mathbf{P} > \mathbf{0}$ ($\mathbf{P} \geq \mathbf{0}$, $\mathbf{P} < \mathbf{0}$ or $\mathbf{P} \leq \mathbf{0}$), we mean that $\mathbf{P}$ is positive definite (positive semi-definite, negative definite or negative semi-definite). The matrix inequality $\mathbf{A} > \mathbf{B}$ means that $\mathbf{A}$ and $\mathbf{B}$ are square Hermitian matrices and that $\mathbf{A} - \mathbf{B}$ is positive definite. A matrix $\mathbf{H} \in \mathbb{C}^{n \times n}$ is Hurwitz (or stable) if all of its eigenvalues have strictly negative real parts. The Kronecker product of matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$ is defined as

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}$$

which satisfies the following properties

$$\begin{aligned} (\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) &= (\mathbf{AC}) \otimes (\mathbf{BD}) \\ (\mathbf{A} \otimes \mathbf{B})^T &= \mathbf{A}^T \otimes \mathbf{B}^T \\ \mathbf{A} \otimes \mathbf{B} + \mathbf{A} \otimes \mathbf{C} &= \mathbf{A} \otimes (\mathbf{B} + \mathbf{C}) \end{aligned}$$

3 Robust consensus based on observer

Consider a group of N identical agents with general linear dynamics. The dynamics of the i th agent are described by

$$\dot{x}_i = \mathbf{A}x_i + \mathbf{B}u_i, \quad y_i = \mathbf{C}x_i \quad (3)$$

where $x_i \in \mathbb{R}^n$ is the agent i 's state, $u_i \in \mathbb{R}^p$ is the agent i 's control input and $y_i \in \mathbb{R}^q$ is the agent i 's output. It is assumed that $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ is stabilisable and detectable.

Assume that each agent has access to the relative output measurement with respect to its neighbours and obtains external disturbances in transmission channels. An observer-type robust consensus protocol is proposed as

$$\begin{aligned} \dot{z}_i &= \mathbf{F}z_i + \mathbf{G} \sum_{j=1}^N a_{ij}(y_j - y_i) + \mathbf{H} \sum_{j=1}^N l_{ij} \mathbf{K} \mathbf{T}_1 \sum_{j=1}^N a_{jm}(y_m - y_j) \\ &\quad + \mathbf{H} \sum_{j=1}^N \mathbf{K} \mathbf{T}_2 (z_j - z_i) + \mathbf{H} \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \\ u_i &= \mathbf{K} \left(\mathbf{T}_1 \sum_{j=1}^N a_{ij}(y_j - y_i) + \mathbf{T}_2 z_i \right) + \omega_i, \quad i = 1, \dots, N \end{aligned} \quad (4)$$

where $\mathbf{F} \in \mathbb{R}^{(n-q) \times (n-q)}$, $\mathbf{G} \in \mathbb{R}^{(n-q) \times q}$, $\mathbf{H} \in \mathbb{R}^{(n-q) \times p}$, $\mathbf{T}_1 \in \mathbb{R}^{n \times q}$, $\mathbf{T}_2 \in \mathbb{R}^{n \times (n-q)}$ and $z_i \in \mathbb{R}^{(n-q)}$ is agent i 's observer state. $\omega_i \in L_2[0, \infty)$ represents external disturbance in transmission channels. The term $\sum_{j=1}^N a_{ij}(y_j - y_i)$ in (4) denotes the information exchanges between agent i and its neighbours. $a_{ii} = 0$ and $a_{ij} = 1$ if agent i can obtain information from agent j , but 0 otherwise. $\mathbf{K} \in \mathbb{R}^{p \times n}$ is the feedback matrix to be determined.

Algorithm 1: Observer: The observer (4) can be constructed as follows:

1. Choose a Hurwitz matrix $\mathbf{F}$ and $\lambda_i(\mathbf{F}) \neq \lambda_j(\mathbf{A})$, $\forall i, j = 1, \dots, n - q$.
2. Choose a matrix $\mathbf{G}$ such that $\{\mathbf{F}, \mathbf{G}\}$ is stabilisable.
3. Solve $\mathbf{TA} - \mathbf{FT} = \mathbf{GC}$ and obtain a non-singular matrix $\mathbf{T}$.
4. $\mathbf{H} = \mathbf{TB}$, $\begin{bmatrix} \mathbf{C} \\ \mathbf{T} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{T}_1 & \mathbf{T}_2 \end{bmatrix}$.

With (4), (3) can be written as

$$\begin{aligned} \dot{\xi}_i &= \Phi \xi_i - \sum_{j=1}^N l_{ij} \Gamma \xi_j + \sum_{j=1}^N l_{ij} \sum_{j=1}^N l_{jm} \Psi \xi_m + \Omega \omega_i \\ &\quad - \sum_{j=1}^N l_{ij} \Upsilon \omega_j, \quad i = 1, \dots, N \end{aligned} \quad (5)$$

where

$$\begin{aligned} \xi_i &= \begin{bmatrix} x_i \\ z_i \end{bmatrix}; \quad \Phi = \begin{bmatrix} \mathbf{A} & \mathbf{BK}\mathbf{T}_2 \\ 0 & \mathbf{F} \end{bmatrix}; \quad \Gamma = \begin{bmatrix} \mathbf{BK}\mathbf{T}_1\mathbf{C} & 0 \\ \mathbf{GC} & \mathbf{HK}\mathbf{T}_2 \end{bmatrix} \\ \Psi &= \begin{bmatrix} 0 & 0 \\ \mathbf{HK}\mathbf{T}_1\mathbf{C} & 0 \end{bmatrix}; \quad \Omega = \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix}; \quad \Upsilon = \begin{bmatrix} 0 \\ \mathbf{H} \end{bmatrix} \end{aligned}$$

l_{ij} is an entry of the Laplacian matrix associated with the communication topology $\mathbf{g}$.

Our objective is to find appropriate feedback matrix $\mathbf{K}$ to suppress external disturbances of agents and make all agents reach agreement. Then, a natural way to combine the relative information is to define a variable $\delta(t)$ computed from the relative disagreements

between agent 1 with the others agents as follows

$$\delta(t) = \begin{bmatrix} \tilde{\xi}_1 \\ \vdots \\ \tilde{\xi}_{N-1} \end{bmatrix} = \begin{bmatrix} \xi_1 - \xi_2 \\ \vdots \\ \xi_1 - \xi_N \end{bmatrix} = (E \otimes I_{2n})\xi(t) \quad (6)$$

where $\tilde{\xi}_i = \xi_1 - \xi_{i+1}$, $i = 1, \dots, N-1$, $\xi(t) = [\xi_1^T, \dots, \xi_N^T]^T$ and $E = [\mathbf{1} - I_{N-1}] \in R^{(N-1) \times N}$. $\delta(t)$ is referred to as the disagreement vector. It can be verified that $\delta(t)$ evolves according to the following so-called disagreement dynamical

$$\dot{\delta} = (I_{N-1} \otimes \Phi - \tilde{L} \otimes \Gamma + \tilde{L}^* \otimes \Psi)\delta + (I_{N-1} \otimes \Omega - \tilde{L} \otimes \Upsilon)\varpi \quad (7)$$

where $\varpi = [\varpi_1^T, \dots, \varpi_{N-1}^T]^T$, $\varpi_i = \omega_1 - \omega_{i+1}$, $i = 1, \dots, N-1$ and $\tilde{L}$ is defined in (2) and $\tilde{L}^* = ELLF = \tilde{L}\tilde{L}$, $F = [\mathbf{0} \quad -I_{N-1}]^T \in R^{N \times (N-1)}$.

Theorem 1: Assume that (A, B, C) is stabilisable and detectable. Systems (5) achieve robust consensus by the protocol (4) under communication topology g if and only if the following $N-1$ subsystems are robust stable

$$\dot{\theta}_i = (A - \lambda_i BK)\theta_i + B\mu_i \quad i = 1, \dots, N-1$$

where λ_i are the eigenvalues of the reduced Laplacian matrix $\tilde{L}$ of g .

Proof: First, according to (6), it can be shown that the robust consensus problem of (5) is equivalent to the robust stability problem of the disagreement dynamics (7).

Let $U \in R^{(N-1) \times (N-1)}$ be a unitary matrix and $\Lambda \in R^{(N-1) \times (N-1)}$ be an upper triangular matrix such that

$$U^{-1}\tilde{L}U = \Lambda$$

where the diagonal entries of Λ are the eigenvalues of $\tilde{L}$. Introduce the state transformation $\varepsilon = (U^{-1} \otimes I_{2n})\delta$ and $\mu = (U^{-1} \otimes I_p)\varpi$, where $\varepsilon = [\varepsilon_1^T, \dots, \varepsilon_{N-1}^T]^T$, $\mu = [\mu_1^T, \dots, \mu_{N-1}^T]^T$. Then, (7) can be represented in terms of ε and μ as follows

$$\dot{\varepsilon} = (I_{N-1} \otimes \Phi - \Lambda \otimes \Gamma + \Lambda \otimes \Psi)\varepsilon + (I_{N-1} \otimes \Omega - \Lambda \otimes \Upsilon)\mu \quad (8)$$

It is easy to see that the elements of the state matrix of (8) are either block diagonal or block upper triangular. Hence, the robust stability problem of system (8) is equivalent to the robust stability of the following $N-1$ isolated subsystems along the diagonal

$$\dot{\varepsilon}_i = (\Phi - \lambda_i \Gamma + \lambda_i^2 \Psi)\varepsilon_i + (\Omega - \lambda_i \Upsilon)\mu_i, \quad i = 1, \dots, N-1 \quad (9)$$

By the similarity transformation

$$\begin{bmatrix} I_n & 0 \\ \lambda_i T & I_n \end{bmatrix}$$

(9) can be converted into the following systems

$$\dot{\eta}_i = \begin{bmatrix} A - \lambda_i BK & BKT_2 \\ 0 & F \end{bmatrix} \eta_i + \begin{bmatrix} B \\ 0 \end{bmatrix} \mu_i, \quad i = 1, \dots, N-1 \quad (10)$$

According to Algorithm 1, the matrix F is Hurwitz. It is easy to see that the robust stability of (10) is equivalent to that of the following

systems

$$\dot{\theta}_i = (A - \lambda_i BK)\theta_i + B\mu_i \quad i = 1, \dots, N-1 \quad (11)$$

Therefore the robust consensus of (5) can be solved if and only if (11) are robust stable.

Remark 1: In this theorem, by the model transformation, it converts the robust consensus problem of complex multi-agent networked systems under the observer-type protocol into the robust stability of a set of isolated subsystems as a single agent, which reduces the computational complexity. Denote the transfer function matrices of (7)–(9) by $T_{\delta\varpi}$, $T_{\varepsilon\mu}$ and $T_{\varepsilon_i\mu_i}$, respectively. Then, it follows that

$$T_{\varepsilon\mu} = \text{diag}(T_{\varepsilon_1\mu_1}, \dots, T_{\varepsilon_{N-1}\mu_{N-1}}) = (U^{-1} \otimes I_{2n})T_{\delta\varpi}(U \otimes I_p)$$

As mentioned above, both $(U^{-1} \otimes I_{2n})$ and $(U \otimes I_p)$ are unitary matrices, so the norm of the $T_{\delta\varpi}$ on (7) is consistent with that of the $T_{\varepsilon\mu}$ on (8). Furthermore, the transformation between (9) and (10) is a similarity transformation that does not change the norm of them too. So, the robust of (7) is consistent with that of (11).

Let $\gamma = \text{Re}_{\min}(\tilde{L})$ denote the smallest real part of the eigenvalues of the reduced Laplacian matrix $\tilde{L}$. For a prescribed scalar ρ , define a performance index by

$$J(\mu_i) = \int_0^\infty \theta_i(t)^T \theta_i(t) - \rho^{-2} \mu_i(t)^T \mu_i(t) dt \quad (12)$$

Assuming that (A, B) is stabilisable, there exists a solution $P > 0$ for the following inequality

$$PA + A^T P + (\rho^2 - 2\gamma)PBB^T P + I_n < 0 \quad (13)$$

Theorem 2: Assume that (A, B, C) is stabilisable and detectable. The invariant communication topology of (5) has a directed spanning tree. Systems (11) can asymptotically achieve robust stability with performance index (12) when $K = B^T P$ where P is defined in (13).

Proof: Assume there is a matrix $P > 0$ and $K = B^T P$. Construct the following Lyapunov function candidate for (11)

$$V(\psi_i) = \psi_i^T P \psi_i, \quad \psi_i = \begin{bmatrix} \theta_i \\ \mu_i \end{bmatrix}$$

Then, the time derivative of this Lyapunov function along the trajectory of (11) is

$$\begin{aligned} \dot{V}(\psi_i) &= \theta_i^T (A^T P + PA - 2\lambda_i PBB^T P)\theta_i + \mu_i^T B^T P \theta_i + \theta_i^T PB \mu_i \\ &\leq \theta_i^T (A^T P + PA - 2\gamma PBB^T P)\theta_i + \mu_i^T B^T P \theta_i + \theta_i^T PB \mu_i \\ &= \begin{bmatrix} \theta_i^T & \mu_i^T \end{bmatrix} \begin{bmatrix} PA + A^T P - 2\gamma PBB^T P & PB \\ B^T P & 0 \end{bmatrix} \begin{bmatrix} \theta_i \\ \mu_i \end{bmatrix} \end{aligned} \quad (14)$$

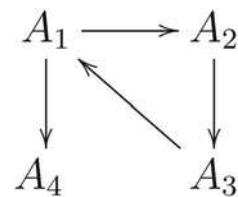


Fig. 1 Communication topology (g)

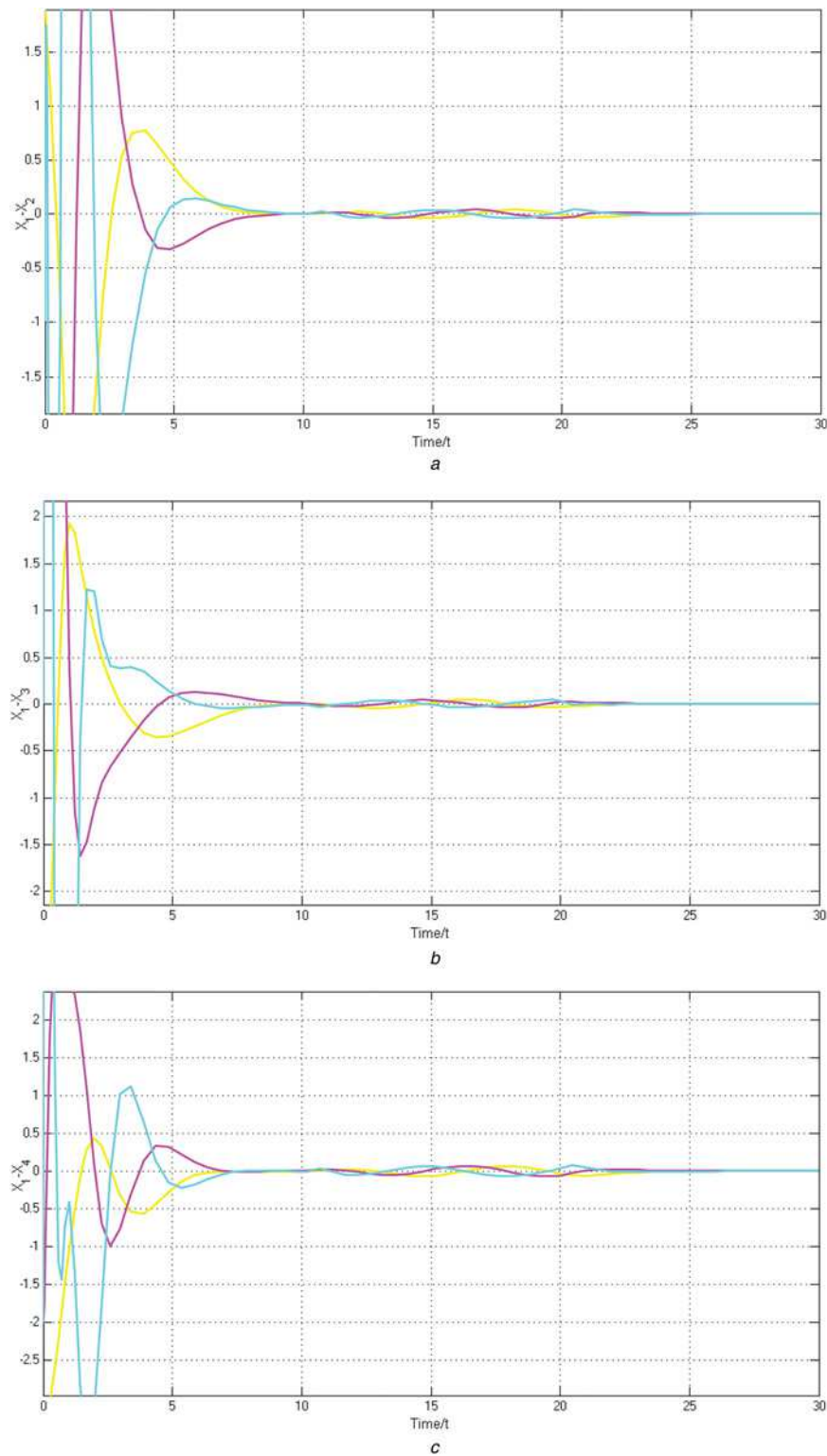


Fig. 2 State error trajectories of the network with topology (Fig. 1)

a $\delta(A_1 - A_2)$

b $\delta(A_1 - A_3)$

c $\delta(A_1 - A_4)$

The performance index (12) can be rewritten as follows

$$J(\mu_i) = \int_0^{\infty} [\theta_i(t)^T \theta_i(t) - \rho^{-2} \mu_i(t)^T \mu_i(t) + \dot{V}(\psi_i)] dt + V(\psi_i)|_{t=0} - V(\psi_i)|_{t=\infty}$$

Since $V(\psi_i)|_{t=0} = 0$ under zero initial condition and $V(\psi_i)|_{t=\infty} \geq 0$, it can be seen that

$$J(\mu_i) \leq \int_0^{\infty} [\theta_i(t)^T \theta_i(t) - \rho^{-2} \mu_i(t)^T \mu_i(t) + \dot{V}(\psi_i)] dt$$

Therefore it can be seen that

$$\begin{bmatrix} \theta_i^T & \mu_i^T \end{bmatrix} \begin{bmatrix} PA + A^T P - 2\gamma PBB^T P + I_n & PB \\ B^T P & -\rho^{-2} I_n \end{bmatrix} \begin{bmatrix} \theta_i \\ \mu_i \end{bmatrix} < 0$$

By taking the schur complement, (13) can be expressed as

$$\begin{bmatrix} PA + A^T P - 2\gamma PBB^T P + I_n & PB \\ B^T P & -\rho^{-2} I_n \end{bmatrix} < 0$$

Hence, the robust stability problem of (11) can be solved.

Remark 2: In this theorem, the performance index of robust stability is converted into an inequality form that is convenient to combine with the Lyapunov inequality. When $K = B^T P$, (11) can achieve robust stability, which also implies that the robust consensus of (5) can be solved. When $\mu_i = 0$, it can be seen that (11) are asymptotically stable if $P > 0$, $K = B^T P$ and

$$PA + A^T P - 2\gamma PBB^T P < 0 \quad (15)$$

Hence, it can be seen that if P can satisfy (13), it must satisfy (15). Since (13) is different from the normal Riccati inequality, some transformation must be applied to solve it. Compared with other robust consensus methods [22], Theorem 2 not only gives necessary and sufficient conditions (13) in terms of linear matrix inequality, but also gives an appropriate feedback matrix $K = B^T P$.

Consider the case where the topology is dynamically switching. Assume the collection of graphs in each interval $[t_k, t_{k+1})$ is jointly connected. There exists a permutation matrix $T_{\sigma(t_k^{j-1})}^T$ in subinterval $[t_k^{j-1}, t_k^j)$ such that

$$T_{\sigma(t_k^{j-1})}^T \tilde{L}_{\sigma(t_k^{j-1})} T_{\sigma(t_k^{j-1})} = \Lambda_{\sigma(t_k^{j-1})}$$

According to Lemma 1, the reduced Laplacian matrix $\tilde{L}_{\sigma(t_k^{j-1})}$ has zero eigenvalues in subintervals $[t_k^{j-1}, t_k^j)$. Let $\gamma' = \text{Re}_{\min}(\tilde{L}_{\sigma(t_k^{j-1})})$ denote the smallest real part of non-zero eigenvalues of reduced Laplacian matrix $\tilde{L}_{\sigma(t_k^{j-1})}$ in interval $[t_k, t_{k+1})$. Assume $\omega_n = 0$ (4) when agent n does not communicate with other agents and so $\mu_i = 0$ (11) when $\lambda_i = 0$. Let $P_2 > 0$ be a solution of the following inequalities

$$\begin{aligned} P_2 A + A^T P_2 &\leq 0 \\ P_2 A + A^T P_2 + (\rho^2 - 2\gamma') P_2 B B^T P_2 + I_n &< 0 \end{aligned} \quad (16)$$

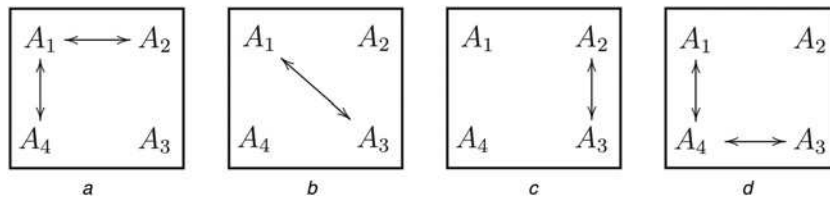


Fig. 3 Jointly connected communication topologies

a g_1
b g_2
c g_3
d g_4

Theorem 3: Assume that (A, B, C) is stabilisable and detectable. The communication topologies of (5) are jointly connected during each interval $[t_k, t_{k+1})$. Systems (11) can asymptotically achieve robust stability with performance index (12) when $K = B^T P_2$ where P_2 is defined in (16).

Proof: Assume there is a matrix $P > 0$ and $K = B^T P$. Construct the following Lyapunov function candidate for (11) in subinterval $[t_k^{j-1}, t_k^j)$

$$V^{j-1}(\psi_i^{j-1}) = \psi_i^{j-1T} P \psi_i^{j-1}, \quad \psi_i^{j-1} = \begin{bmatrix} \theta_i^{j-1} \\ \mu_i^{j-1} \end{bmatrix}$$

Since the topologies are jointly connected, it can be verified that every λ_i will be greater than zero at least once in subintervals of each interval $[t_k, t_{k+1})$. Then, the time derivative of this Lyapunov function along the trajectory of (11) in subinterval $[t_k^{j-1}, t_k^j)$ is

$$\begin{aligned} \dot{V}^{j-1}(\psi_i^{j-1}) &= \theta_i^{j-1T} (A^T P + PA - 2\lambda_i PBB^T P) \theta_i^{j-1} \\ &\quad + \mu_i^{j-1T} B^T P \theta_i^{j-1} + \theta_i^{j-1T} PB \mu_i^{j-1} \\ &\leq \theta_i^{j-1T} (A^T P + PA - 2\gamma' PBB^T P) \theta_i^{j-1} \\ &\quad + \mu_i^{j-1T} B^T P \theta_i^{j-1} + \theta_i^{j-1T} PB \mu_i^{j-1} \quad \lambda_i > 0 \\ \dot{V}^{j-1}(\psi_i^{j-1}) &= \theta_i^{j-1T} (A^T P + PA) \theta_i^{j-1} \quad \lambda_i = 0 \end{aligned}$$

According to the result of Theorem 2, it can be concluded that when (16) is satisfied and $K = B^T P_2$, (11) can asymptotically achieve robust stability with performance index (12) under jointly connected communication topologies.

Remark 3: The jointly connected communication topologies will ensure that every λ_i will be greater than zero at least once in subintervals of each interval $[t_k, t_{k+1})$. From the view of stability, (16) will make the Lyapunov function $\dot{V}^{j-1}(\psi_i^{j-1}) < 0$, in some subintervals with $\lambda_i > 0$ and make the Lyapunov function $\dot{V}^{j-1}(\psi_i^{j-1}) \leq 0$, in some subintervals with $\lambda_i = 0$. So it can be seen that in each interval $[t_k, t_{k+1})$, the total Lyapunov function $\dot{V}(\psi_i) < 0$ will ensure the robust stability of (11). Furthermore, if the union of the communication topologies is not jointly connected after some finite time, robust consensus cannot be achieved.

4 Simulation results

In this section, numerical simulations will be given to illustrate the theoretical result obtained in the previous section.

Consider the following multi-agent systems consisting of four agents with external disturbances in transmission channels under

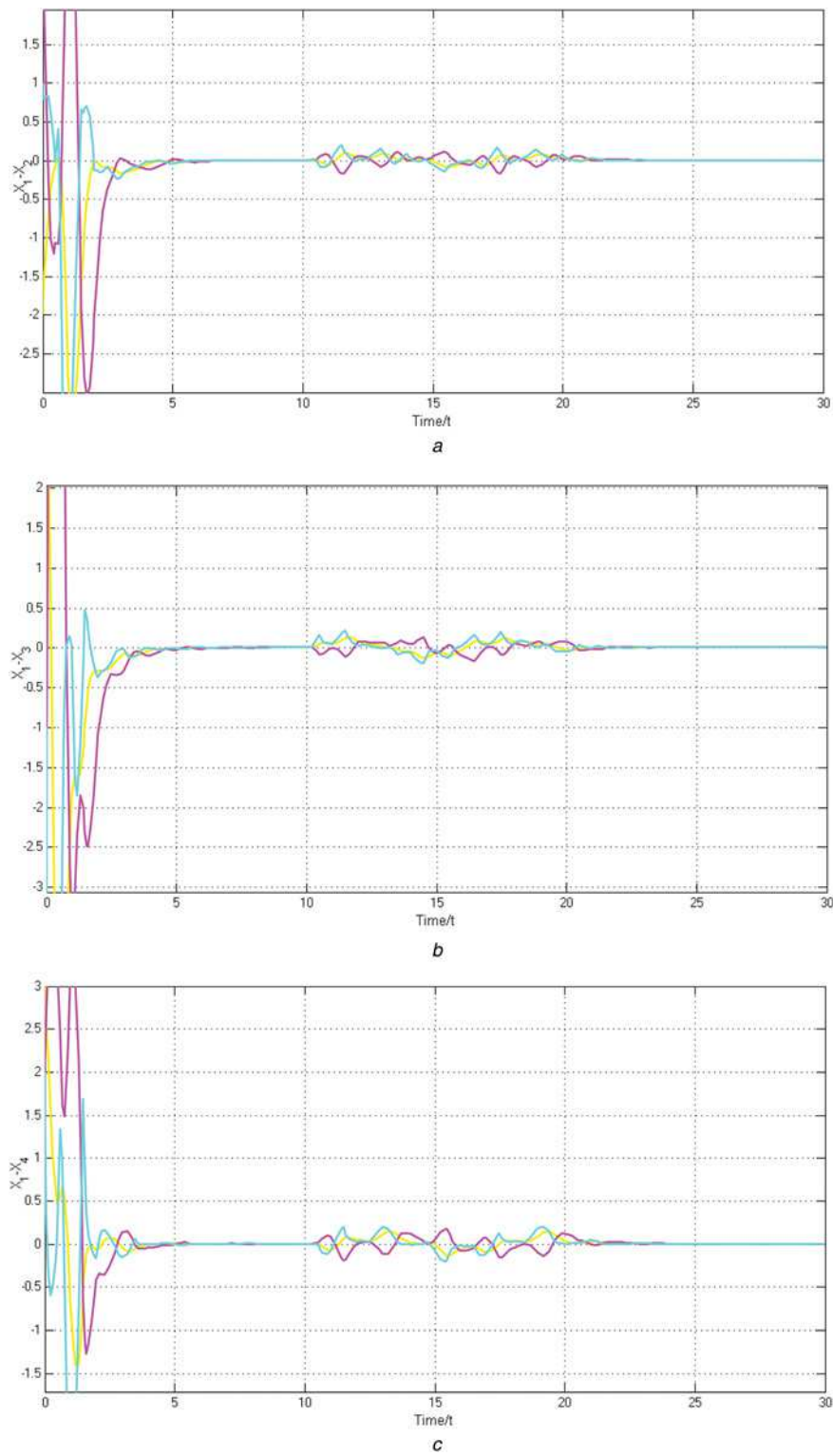


Fig. 4 State error trajectories of the network with switching topologies (Fig. 3)

a $\delta(A_1 - A_2)$

b $\delta(A_1 - A_3)$

c $\delta(A_1 - A_4)$

a time invariant communication topology. The system matrices are

It is easy to verify that (A, B, C) is stabilisable detectable. According to Algorithm 1, we can construct an observer as follows

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$F = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad H = \begin{bmatrix} 7 \\ 128 \\ 3 \\ -32 \end{bmatrix}$$

$$T = \begin{bmatrix} \frac{1}{8} & -\frac{3}{32} & \frac{7}{128} \\ 0 & \frac{1}{8} & -\frac{3}{32} \end{bmatrix}$$

The communication topology g is described in Fig. 1.

Then, the Laplacian matrix L of communication topology g is as follows

$$L = \begin{bmatrix} 1 & 0 & -1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

Using MATLAB, we can obtain the eigenvalues: $1, 1.5 \pm i0.866$ of $\tilde{L}$ and so $\gamma = 1$. When $\rho = 0.2$, a solution P of (13) and $K = B^T P$ as follows

$$P = \begin{bmatrix} 5.1042 & 5.6917 & 2.1136 \\ 5.6917 & 10.4292 & 4.3442 \\ 2.1136 & 4.3442 & 3.0764 \end{bmatrix}$$

$$K = [2.1136 \quad 4.3442 \quad 3.0764]$$

We take

$$\omega = \begin{cases} 0, & t \in [0, 10) \\ w(t), & t \in [10, 20) \\ 0, & t \in [20, 30) \end{cases} \quad (17)$$

where $w(t) = [\omega_1, \omega_2, \omega_3, \omega_4]^T = [0.1 \sin(t), 0.2 \cos(t), -0.3 \sin(t), 0.4 \cos(t)]^T$. The state error trajectories between the agents A1 and A2, A3, A4 under the protocol (4) are shown in Fig. 2. It can be seen from Fig. 2 that when $t \in [0, 10)$, four agents can achieve consensus; when $t \in [10, 20)$, the protocol (4) can efficiently suppress the external disturbances and make four agents achieve robust consensus; and when $t \in [20, 30)$, the four agents can achieve consensus again which validates our proposed method.

Consider the following multi-agent systems consisting of four agents with external disturbances in transmission channels under jointly connected communication topologies. The system matrices are

$$A = \begin{bmatrix} -2.9967 & -0.0488 & 2.8304 \\ 7.9832 & -2.8765 & -7.2792 \\ -7.4749 & 2.7716 & 5.1233 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = [1 \quad 0 \quad 0]$$

It is easy to verify that (A, B, C) is stabilisable and detectable. According to Algorithm 1, we can construct an observer as follows

$$F = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 0.1007 & 0.0993 & 0.5876 \\ 0.05693 & -0.1295 & -0.1415 \end{bmatrix}$$

$$H = \begin{bmatrix} 0.5876 \\ -0.1415 \end{bmatrix}$$

The possible communication topologies are described in Fig. 3.

Using MATLAB, we can obtain $\gamma' = 1$. When $\rho = 0.2$, a solution P_2 of (16) and $K = B^T P_2$ as follows

$$P = \begin{bmatrix} 9.5067 & -1.8191 & -3.2484 \\ -1.8191 & 2.2441 & 0.9861 \\ -3.2484 & 0.9861 & 6.7513 \end{bmatrix}$$

$$K = [-3.2484 \quad 0.9861 \quad 6.7513]$$

The communication topologies are switched as $g_1 \rightarrow g_2 \rightarrow g_3 \rightarrow g_4 \rightarrow g_1 \rightarrow \dots$. It is easy to see that the collection of $\{g_1, g_2\}$ and the collection of $\{g_3, g_4\}$ are jointly connected. Every subinterval is active for 0.25 s. We take ω defined in (17). The state error trajectories between agents A1 and A2, A3, A4 under the protocol (4) are shown in Fig. 4. It can be seen that the four agents can achieve robust consensus with external disturbances under jointly connected communication topologies.

Remark 4: In this example, system matrix A has one negative real and two purely imaginary eigenvalues. This means the value of Lyapunov function of an agent without external input will not reduce to zero. So it can be seen that the jointly connected communication topology is indispensable for achieving robust consensus. Although the switching frequency does not affect the robust consensus of systems when a jointly connected communication topology and (16) are satisfied, it can affect the convergence process of systems.

5 Conclusions

In this paper, robust consensus of linear multi-agent systems with external disturbances in transmission channels under fixed and switching communication topologies was considered. A distributed observer-type robust consensus protocol based on relative output measurement was proposed. A model transformation approach was introduced to analyse the robust consensus of multi-agent systems. Necessary and sufficient conditions for robust consensus were given in terms of linear matrix inequalities. However, we did not consider the issue of system model uncertainty, time-delay or adaptive consensus in this paper. Hence, these issues will be investigated in future work.

6 References

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